



Tipping Cancer Cells Over the Edge: The Context-Dependent Cost of High Ploidy

Noemi Andor¹, Philipp M. Altrock^{1,2}, Navami Jain^{1,3}, and Ana P. Gomes⁴

ABSTRACT

Tetraploidy is an aneuploidy-permissive condition that can fuel tumorigenesis. The tip-over hypothesis of cytotoxic therapy sensitivity proposes that therapy is effective if it pushes a cell's aneuploidy above a viable tipping point. But elevated aneuploidy alone may not account for this tipping point. Tissue microenvironments that lack sufficient resources to support tetraploid cells

can explain the fitness cost of aneuploidy. Raw materials needed to generate deoxynucleotides, the building blocks of DNA, are candidate rate-limiting factors for the evolution of high-ploidy cancer cells. Understanding the resource cost of high ploidy is key to uncover its therapeutic vulnerabilities across tissue sites with versatile energy supplies.

Introduction

Analyses of intratumor heterogeneity across multiple cancer types suggest that tumor cell fitness declines once the load of somatic copy-number alterations (SCNA) exceeds a certain limit (1, 2). Multiple studies across an unexpectedly wide range of cancer types consistently confirmed the amount of SCNAs that define this limit at approximately 75% of the genome (1, 3, 4). Tumors that exceed the limit are associated with a better outcome than those below the limit. However, this difference is only significant among therapy-naïve patients, not among patients who subsequently underwent cytotoxic therapy (Fig. 1A; refs. 1, 5, 6). The cause for this remains unknown.

Two hypotheses can be formulated to explain the association between SCNA load and risk of disease progression in the two patient strata. We previously put forward the “tip-over hypothesis of DNA damage therapy sensitivity” (7). This hypothesis proposes that cytotoxic therapy, when administered to tumors close to the SCNA limit, pushes SCNA burden beyond the limit (7), thereby causing a net decrease in tumor population fitness. Other recent work however supports an alternative hypothesis (8, 9): therapy causes a net increase in fitness of tumors that exceed the limit (Fig. 1). Hereby the tipping point is accounted for not by elevated SCNA burden, but by an inability of the tissue microenvironment (TME) to provide resources needed to evolve and maintain high SCNA burden.

High SCNA burden accompanies high ploidy (Fig. 1B). Cell populations with more DNA are often observed further apart from euploid states (10, 11). High ploidy serves as a buffer against deleterious effects of genetic alterations and facilitates tolerance of SCNAs (10–12). A high-ploidy cell population might explore the adaptive fitness landscape more easily. Hereby, an adaptive fitness landscape is a genotype-fitness map, which associates to each SCNA

state a fitness value. Despite the ability of high-ploidy populations to survive and surpass SCNA states located at steep fitness valleys, *in vivo* ploidy does not reach the high levels observed *in vitro* (8, 10, 13). Potential explanations for this discrepancy include immune system interactions, or that stronger energetic limitations govern *in vivo* tumor growth (Fig. 1C).

The vast majority of human cells are diploid (i.e., they carry 46 chromosomes). A whole-genome duplication (WGD) transforms a diploid cell into tetraploid one and has been shown to occur in 28%–56% of human cancers (14–16). Hyperdiploid, near-triploid, and tetraploid cells (further referred to as high-ploidy cells) need to replicate increasing amounts of DNA. The building blocks of DNA are nucleotides, and therefore nucleotide concentrations are central for a cell's ability to replicate DNA. Nucleotide synthesis, increasingly pivotal for high-ploidy cells, is an energy intensive anabolic process that uses multiple metabolic pathways across different cell compartments and several sources of carbon and nitrogen.

An important group of nucleotides are deoxyribose nucleoside triphosphates (dNTP). To maintain homeostasis, dividing cells need to replenish dNTPs at the same rate as consumed by DNA replication. Thus, cell-cycle progression is tightly linked to the ability of cells to acquire nutrients, generate metabolic energy, and to drive anabolic pathways. A salvage pathway exists that recycles free nitrogen bases and nucleosides arising from nucleic acid breakdown and diet (17). However, this pathway is not sufficient for dNTP level maintenance in the majority of highly proliferating cells. These cells need to synthesize dNTPs *de novo* from simple building blocks (glucose, glutamine, aspartate, CO₂, and phosphate donors) to meet their steep need to synthesize nucleic acids. Nucleotide synthesis is a result of tight coordination between the supply of its building blocks, the concentration and activities of all the enzymes in the pathways, and the regulatory mechanisms that control these metabolic networks (Fig. 2A). This regulation is particularly important in the context of cancer (Fig. 2B). Tumorigenesis is often accompanied by cellular metabolic reprogramming that enables cancer cells to adapt and sustain their energetic demands (18, 19). Upregulation of nucleotide synthesis was recently shown to be sufficient to drive oncogenic transformation (20).

In this perspective, we discuss potential selective pressures exerted by the TME by considering the levels of dNTPs, their building blocks, and coexistence of cancer cell populations with variable ploidy. While we primarily focus on DNA replication rate as a measure of cell fitness, we conclude with a broader view on the phenotypic consequences of high ploidy, including invasive potential and persistence.

¹Department of Integrated Mathematical Oncology, H. Lee Moffitt Cancer Center & Research Institute, Tampa, Florida. ²Department of Evolutionary Theory, Max Planck Institute for Evolutionary Biology, Ploen, Germany. ³Stanford University, Stanford, California. ⁴Department of Molecular Oncology, H. Lee Moffitt Cancer Center & Research Institute, Tampa, Florida.

Corresponding Author: Noemi Andor, Integrated Mathematical Oncology, Moffitt Cancer Center, 12902 Magnolia Drive, Tampa, FL 33612. E-mail: noemi.andor@gmail.com

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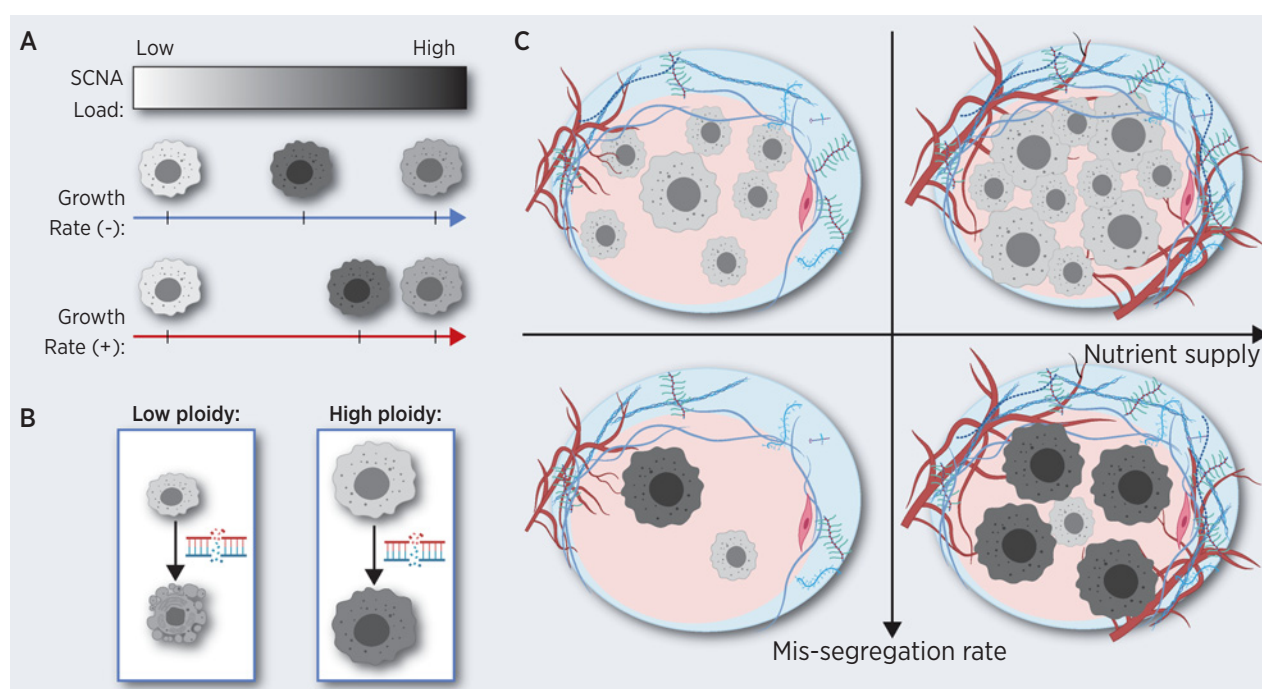


Figure 1.

Opposing selective forces explain why cancer therapies shift the Goldilocks zone of tumor aneuploidy. We distinguish cells along two dimensions: their load of SCNAs (grayscale) and their ploidy (drawn as cell size). **A**, The Goldilocks zone of tumor aneuploidy lies at intermediate SCNA loads among therapy-naïve patients (–; blue arrow) and patients exposed to chemotherapy/radiotherapy (+; red arrow; refs. 1, 3, 4). **B** and **C**, Interaction between two selective pressures—energy scarcity and chromosomal instability—can explain differences between therapy-naïve and therapy-exposed patients. Compared with low-ploidy cells, high-ploidy cells are more likely to survive mutations and accumulate high SCNA loads (**B** and y-axis in **C**). High-ploidy cells require more nutrients for growth (x-axis), setting high-ploidy cells at a disadvantage when competing against low-ploidy cells in nutrient-scarce microenvironments (**C**). The dynamics change when low- and high-ploidy cells compete in mutagenic conditions (y-axis), for example, imposed by cytotoxic therapies.

Materials and Methods

S-phase duration as a function of ploidy and dNTP

We used previously reported Michaelis-Menten kinetics for the average DNA replication rate of polymerase (21), to predict at what dNTP concentrations S-phase duration of high-ploidy cell lines diverges from that of low-ploidy cells (**Fig. 3B**). Let n_p be the number of polymerases active in a cell of ploidy p , then:

$$S_{dna} = n_p \cdot v_{poly} \cdot \frac{S_{dNTP}}{S_{dNTP} + p \cdot k_{poly}} \quad (A)$$

is the number of base pairs of DNA produced by the polymerases in 1 hour. Hereby S_{dNTP} is the intracellular concentration of dNTP (the substrate of polymerase). v_{poly} and k_{poly} are the maximum DNA synthesis rate of a single polymerase and the dNTP concentration required for half-maximum rate, respectively. By assuming that both the number of polymerases expressed in a cell and the dNTP resources they demand are proportional to ploidy, we obtain increasingly divergent DNA synthesis rates between low- and high-ploidy cells as dNTP concentrations decrease (**Fig. 3B**).

S-phase duration as a function of ploidy and O_2

We consider the rate-limiting step of dNDP synthesis—the reduction of ribonucleoside diphosphates via RNR:

$$S_{dNDP} = v_{mr} \cdot \frac{S_{O_2}}{S_{O_2} + p \cdot k_{mr}}, \quad (B)$$

where v_{mr} and k_{mr} were estimated from ref. 53. Here we also assume substrate requirements for RNR increase with increasing ploidy p . To model how DNA synthesis rate depends on O_2 , we neglect the phosphorylation step ($dNDP \rightarrow dNTP$) and use S_{dNDP} as input to Eq. A (i.e., we assume $dNDP \cong dNTP$). We note that in contrast to the polymerase case (Eq. A), here we do not assume that RNR expression increases with increasing ploidy.

Availability of dNTP Affects Tumor Evolution and Cancer Cells Ploidy

Tetraploid clones expand early in tumor evolution—an observation going back to a landmark study in Barrett's esophagus published by Galipeau and colleagues (22) that has since been confirmed for several other tumor types (15, 23, 24). These tumors later lose parts of their tetraploid genome to converge on a highly favorable and commonly observed near-triploid karyotype (12, 25, 26). WGD represents a more viable path toward this favorable near-triploid karyotype, as the resulting tetraploid intermediate can explore the genotype space more effectively in search for fitter cell states (23, 27). If a tetraploid subclone expands, it consistently does so early in tumor evolution (15, 23), when cell density is still low and competition for nutrients is comparatively weak, suggesting these conditions may favor the growth of tetraploid cells. In a model that accounts for the increased resource demands of high-ploidy cells, the availability of cell-extrinsic resources has been suggested to influence coevolution between high- and low-ploidy states (9).

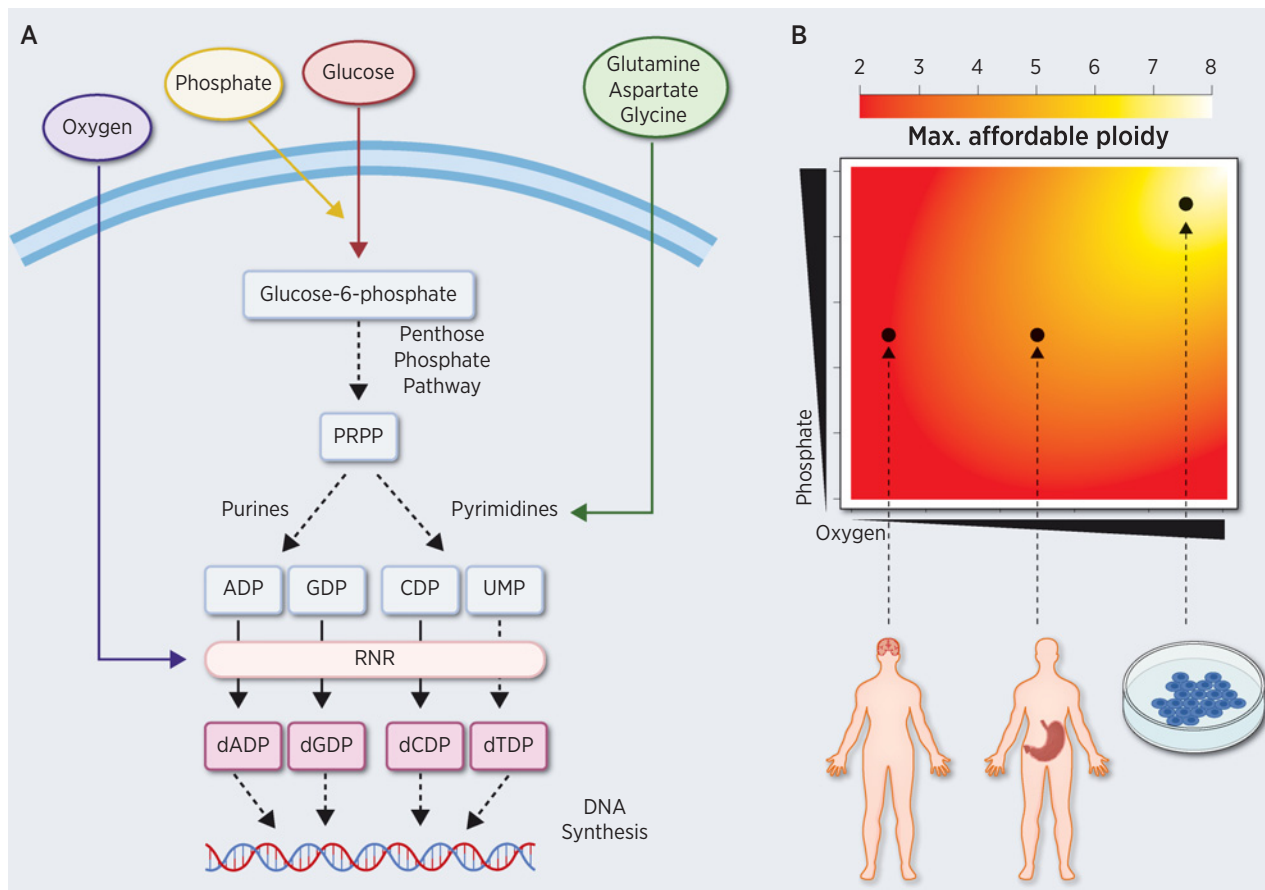


Figure 2.

Availability of dNTP building blocks facilitates the evolution of high-ploidy cancers. **A**, dNTP synthesis relies on extracellular components, including oxygen, phosphate, and glucose. **B**, We hypothesize that the maximum DNA a cell can afford is a function of PO_4 and O_2 saturation, specifically of the minimum between the two. TMEs that have low concentration of either of these resources cap evolution of ploidy at near-diploid states. The relative locations of brain TMEs, stomach TMEs and *in vitro* environments are hypothesized in the oxygen-phosphate phase diagram. *In vitro* populations can afford very high ploidy.

Cell-extrinsic resources are needed to synthesize dNTPs, and dNTP synthesis is at the nexus of cell fate decisions of cancer cells. If dNTP synthesis is slow, it will decelerate replication forks and extend S-phase duration. It has been generally thought that tumor cells have very similar duration of S-phase, but recent evidence points to the contrary, emphasizing variability of S-phase duration in tumors (28). Prolonged S-phase increases the risk of replication stress and cell death. Cells that spend more than 20 hours in S-phase have an increased risk of dying upon entering M-phase, compared to cells that spend less than 20 hours in S-phase (29). High-ploidy cells need to cope with increased amounts of DNA during replication to maintain their fitness.

DNA replication relies on polymerase epsilon and delta to assemble dNTPs into a copy of the genome (30, 31). The synthesis rate of dNTPs is proportional to the amount of active polymerases (32), and ploidy can serve as proxy of polymerase activity (Fig. 3A). This proximal relation suggests that S-phase duration is relatively constant despite variability in the amount of nuclear DNA, provided the building blocks of dNTP synthesis are readily available. In conjunction with Michaelis-Menten kinetics for the average DNA replication rate of a single polymerase (21), the number of active polymerases per cell can thus be used to predict the critical dNTP concentration at which S-phase duration diverges with ploidy (Fig. 3B; Materials and Methods). Taken

together, these insights suggest that limiting dNTP concentrations explain why high-ploidy cells experience more replication stress than low-ploidy cells (33). Considering that replicating cells have been shown to increase dNTP synthesis by 22- to 320-fold compared with nondividing cells (34), we pose that, to maintain fitness, high-ploidy cells need to be able to *de novo* synthesize sufficient dNTPs to replicate their increased DNA content.

Phosphate as a Rate-Limiting Factor for High-ploidy Cells

Near-triploid karyotypes are common in gastrointestinal cancers, where incidence of WGD is 52% (10). Interestingly, gastrointestinal cancer cells *in vitro* often evolve to even higher ploidy than the near-triploid states observed in patients, raising the idea that the gastrointestinal environment curbs the cell population's ploidy level (Fig. 4C).

Nucleic acids contain 9% phosphate (PO_4) per unit dry mass, which is considerably higher than PO_4 levels in most other biomolecules (35). Normal serum PO_4 concentration follows the circadian rhythm and ranges from 2.5 to 4.5 mg/dL—its homeostasis is largely maintained by the kidneys and the small intestine. Thirty percent of gastrointestinal

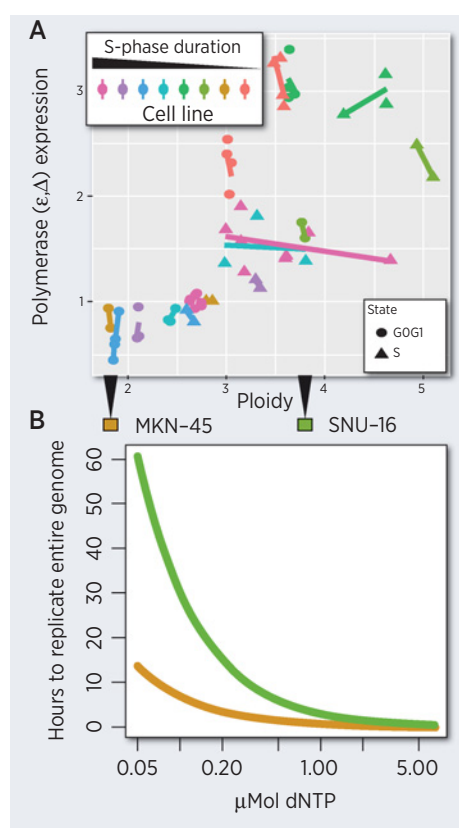


Figure 3.

S-phase duration and ploidy in stomach cancer cell lines. **A**, High-ploidy karyotype clones (dots) across nine stomach cancer cell lines (color-code; ref. 1) express higher levels of polymerase epsilon and delta (y-axis). Correlation observed among both S-phase (Pearson $r = 0.54$; $P = 0.004$) and G_0 - G_1 representatives ($r = 0.82$, $P = 2E-8$) of a given clone. Cell lines are sorted according to the duration of their S-phase (no correlation between S-phase duration and ploidy was observed; $r = -0.08$; $P = 0.54$). **B**, Using each cell line's ploidy as proxy of how many polymerases are replicating its DNA, we project how time required to complete S-phase (y-axis) increases with decreasing dNTP concentrations. Although at saturation, S-phase duration of MKN-45 and SNU-16 is similar, we predict that at limiting dNTP concentrations, SNU-16 (high-ploidy cell line) will take 10.3 hours longer to replicate its DNA than MKN-45 (low-ploidy cell line).

PO_4 transport is dependent on vitamin D (36), and vitamin D deficiency can directly result from malignancy or cancer treatment (36). Disordered regulation of phosphate, as a consequence of neoplasia is relatively common and is known as oncological hypophosphatemia (37). Cellular sequestration of PO_4 by rapidly replicating tumor cells can cause low-serum PO_4 concentrations (37). Grade 3 hypophosphatemia has been reported in 27.5% of stomach cancers, 27% of esophageal cancers, and 18% of colon cancers (38). Interstitial inorganic phosphate was also found to be several folds higher in breast cancer tumors compared with normal mammary glands and in metastatic compared with non-metastatic tumors (39). None of these findings could be explained by a difference in the degree of vascularization. A 2-fold increase in PO_4 concentrations was also found in colon and lung tumors compared with surrounding healthy tissues (40). In contrast, kidney and liver cancers showed no differences in PO_4 levels when compared with normal kidney and liver tissue (40).

Interestingly, WGDs are also more frequently observed in colon and lung cancers than in kidney and liver cancers (15, 41). These tissue-specific differences could be used as a starting point to gain mechanistic insights in the conditions that select for high-ploidy cells.

Haploid yeast cells are better competitors than diploids in low PO_4 environments (42–45). A study analyzing changes in phosphate metabolism during the cell cycle of yeast found that consumption of extracellular orthophosphate increased by 14 mmol per liter during DNA synthesis (46). Extrapolating to the size of the human genome, we can derive a hypothetical form of a cell's S-phase duration as a function of its ploidy and extracellular PO_4 concentration (Fig. 4C). This analysis indicates that tetraploid populations require approximately 200 mmol/L more PO_4 to replicate their genomes in 20 hours, compared with requirements for replication of triploid genomes. While PO_4 concentrations may allow for the replication of triploid genomes in reasonable timeframes, they are unlikely to meet the demands of tetraploid genomes. Taken together, these studies give rise to the idea that for several cancer types, including gastrointestinal cancers, ploidy levels are a function of PO_4 availability within the tissue they inhabit.

Oxygen Tension Conditions the Evolution of High-Ploidy Cancer Cells

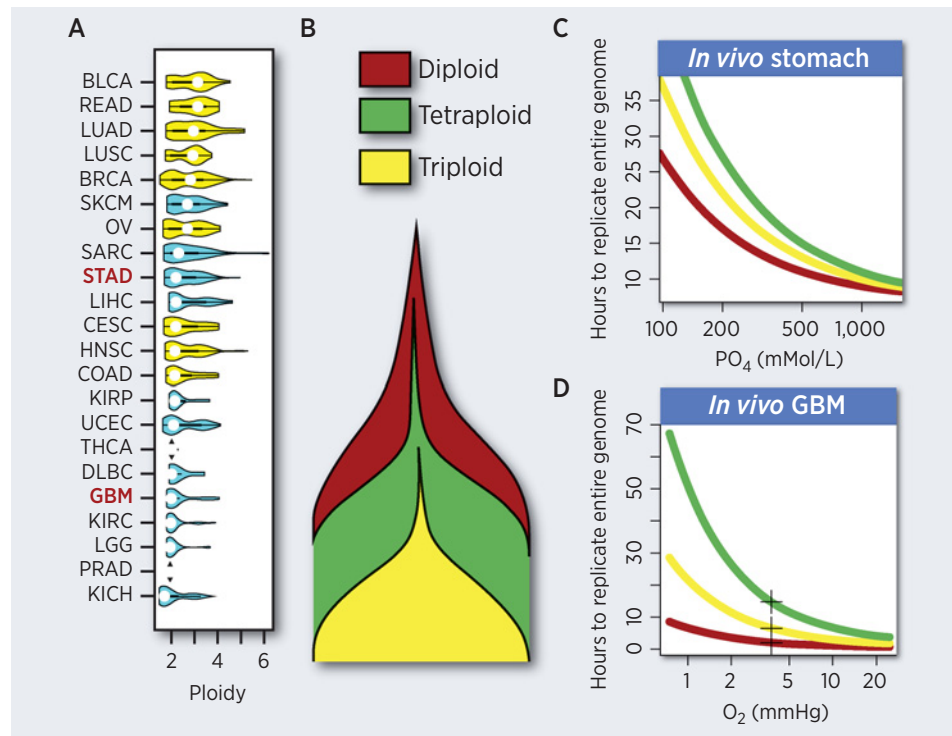
Cancers differ significantly in their ploidy levels. Rates of WGD range from less than 10% in non-Hodgkins lymphoma to almost 60% in germ cell tumors (15). We would expect similar differences in PO_4 levels across tissue types, if tissue specificity of PO_4 levels alone determined evolvability of high ploidy. Yet, PO_4 levels are similar across most normal tissue types (47, 48), suggesting that, in a subset of cancers, shortages of other dNTP building blocks limit the evolution of high ploidy before PO_4 becomes scarce. Brain cancers, such as glioblastoma multiforme (GBM), are among the cancer types with the lowest incidence of WGD (Fig. 4A; refs. 10, 15). Even though, oxygen saturation does not fall below 4% in most tissue types, in brain tissue oxygen levels often go down to 0.5% (49).

dNTP synthesis also depends on oxygen availability (49, 50). The rate-limiting step of dNTP synthesis is reduction of ribonucleoside diphosphates via ribonucleotide reductase (RNR; Fig. 2A; ref. 51). Oxygen is essential for mammalian RNR function, but under hypoxic conditions, RNR is able to maintain low levels of dNTP formation by switching subunits (52). Nevertheless, dNTP synthesis is 75% more efficient when RNR functions in the aerobic mode (52). We used Michaelis-Menten kinetics of RNR activity as a function of oxygen (53) to derive a hypothetical form of a cell's S-phase duration as a function of ploidy and O_2 pressure (Fig. 4B and D, Materials and Methods). This analysis indicates that at the lower end of physiologic cerebral oxygen concentrations (49), tetraploid cells would need more than twice as long to complete S-phase compared with triploids (Fig. 4D). Tetraploid cells replicating in the brain may therefore experience a major disadvantage when competing with low-ploidy cells, even before the tumorigenic process itself amplifies oxygen scarcity. Oxygen levels lie at the intersection of several phenomena that are unique to GBM. Stark gender differences in GBM pathology coincide with the sexual dimorphism in physiologic oxygen perfusion levels in the brain (54, 55). Pseudopalisading necrosis in GBM cooccurs near steep oxygen gradients (56, 57).

Together, these studies support the idea that triploid and tetraploid states can often be a disadvantage in the brain, as they demand dNTP synthesis rates that cancer cells cannot sustain due to inherent low

Figure 4.

Predicting how availability of dNTP building blocks impacts S-phase duration in brain and stomach TMEs. **A**, Ploidy distribution across cancer types. High ploidy is frequently observed in stomach cancers, but rarely in GBM (both highlighted red). Cancers where dNTP synthesis inhibitors are part of standard of care are highlighted yellow. Cancer types are ordered from highest to lowest ploidy. Ploidy estimates from a PAN-cancer analysis of whole genomes (10) were used here. **B**, Changes in tumor clonal composition during progression. Tetraploid clones expand early in tumor evolution. These tumors later lose parts of their tetraploid genome to converge on a near-triploid karyotype. **C** and **D**, Ploidy informs how the time required for a cell to replicate its genome increases with decreasing PO_4 (**C**) and O_2 (**D**) concentrations. Crosses indicate physiologic cerebral oxygen concentrations (49).



abundance of O_2 in brain tissue. Without the capacity to sustain double the amount of DNA, these tumors are limited in their evolvability and may be trapped at local fitness optima. This may explain why the vast majority of GBM fall into a single, well-defined karyotype defined by chromosome 7 amplification and chromosome 10 deletion, whereas stomach cancers have highly variable karyotypes (10). Together, these findings suggest that oxygen availability in the environment plays a key role in conditioning the evolution of high-ploidy cancer cell populations.

Carbon Availability Is Unlikely to Limit dNTP Synthesis for Replication of High-Ploidy Cells

The skeleton of dNTPs is derived from carbon, in the form of glucose (17). Therefore, the availability of glucose is also critical for dNTP synthesis and consequently affects the ability to replicate high-ploidy genomes. Metabolic reprogramming that relies on high rates of glucose uptake to enable anabolism and the production of building blocks for macromolecule synthesis, such as dNTPs, is a well-known hallmark of cancer (58, 59). Metabolic stress selects for SCNA signatures predictive of glycolytic phenotypes, and coordinate amplification of glycolytic genes are recurrent events in human tumor evolution (60). This pattern underscores the importance of carbon sources and metabolic substrates for the evolution of high ploidy. Thus, an important question arises: does the availability of carbon in the host organ condition the evolution of high-ploidy cells?

The liver, in many ways the metabolic hub of the human body, removes about two-thirds of glucose and the majority of dietary amino acids from the blood, which are then used to produce other fuel sources for other peripheral organs (61). Thus, liver cells have ample access to nutrients including glucose and glutamine. In support of the idea that

replication of high-ploidy cells is dependent on the metabolic status of the tissue, hepatocytes, the main cell type in the liver, are among the very few somatic cell types that display a certain degree of polyploidy in physiologic conditions (62, 63). Importantly, hyperglycemia that occurs as a consequence of diabetes is known to significantly increase the ability of many cancer types to proliferate, progress into metastatic disease and overcome chemotherapies (64). Interestingly, studies examining context-dependent fitness effects of ploidy in *Saccharomyces cerevisiae* found diploids to have no growth disadvantage over haploids in glucose-limited media (42, 45), suggesting that availability of glucose in the environment may not be rate limiting for the replication of high-ploidy genomes. These studies from yeast, combined with the ability of the liver to maintain glucose homeostasis through gluconeogenesis (61), suggest that carbon shortages at a systemic or local organ level are unlikely to slow down dNTP synthesis to a degree that can condition the evolution of high-ploidy cancer cells.

dNTP Synthesis and Response to Therapy

Previous studies have shown that clones that appear early in tumor evolution have a higher chance to persist than fitter clones who emerge later (65). While tetraploid cells may persist long after increasing nutrient scarcity renders them at a disadvantage, they are likely to emerge within a narrow time window, namely during population bottlenecks. In addition to precancerous states and early stages in tumor evolution, therapy administration constitutes such a bottleneck. Multiple studies have described a minority population of high-ploidy cells, also known as polyan euploid cancer cells, with an unusual resilience to stress (66–68). The presence of greater amounts of genetic material serves as a backup system to protect against deleterious effects of genetic alterations, including those caused by

DNA-damaging therapies (10–12). Kuznetsova and colleagues have shown that tetraploid RPE-1 cells are more resistant to chemotherapeutic intervention than diploid RPE-1 cells (69). High ploidy, although energetically costly, may thus provide evolutionary rescue to a tumor that consists of otherwise therapy-sensitive, low-ploidy clones. When therapy is removed, evolution again favors descendants of high-ploidy cells with less DNA (25, 70), giving rise to a heterogeneous population.

Variability in the availability of raw materials for dNTPs across organs and patients suggests that the TME of these tumors influences the efficacy of cytotoxic therapies. TMEs in which concentrations of certain dNTP building blocks are naturally low, like GBMs, may not support the evolution of high-ploidy cells, thereby depriving the tumor of this particular escape mechanism. In contrast, TMEs with high concentrations of dNTP building blocks, like gastrointestinal and lung environments, are likely to favor the evolution of high-ploidy cells as a route to therapy resistance.

Fluorouracil and gemcitabine are two pyrimidine analogs commonly used for the treatment of certain solid cancers (71). By blocking dNTP production, these agents mimic environments with low availability of dNTP building blocks (72). Interestingly, fluorouracil and gemcitabine are important agents for testicular, ovarian, colorectal, breast, esophageal, head and neck, cervical, bladder and non-small cell lung cancer treatments. Notably, seven of these nine cancers are within the upper 50% of high-ploidy cancers (Fig. 4A; ref. 15). Thus, the efficacy of these agents, in particular when combined with DNA-damaging agents (73), may be in part attributable to a heightened sensitivity of the high-ploidy compartment within these cancer types. Together, these insights suggest that the energetic costs of high ploidy are important factors to consider for forecasting therapy resistance and designing combination therapies that aim to delay resistance.

Conclusions and Future Perspectives

Cancers are multicellular ecosystems. Interactions between tumor cells and their environment manifest as extensive spatiotemporal heterogeneity. One perspective on tumor cell heterogeneity is variability in the ploidy of coexisting tumor cells. High ploidy equips tumors with resilience to cytotoxic therapies. But how does the TME select for the ploidy of cancer cells? Here, we propose that differences in resource availability for dNTP synthesis in the environment are important factors that limit the evolution of high-ploidy cells. While we focused on gastrointestinal and brain environments, this mechanism should also apply to several other cancer types and may explain the variable ploidy levels seen across different primary and metastatic sites (Fig. 4A). For example, primary breast tumors display a predilection for near-diploid karyotypes. However, metastases derived from the same tumors were found to be enriched for cells with near-triploid karyotypes (74). Observations in colo-

rectal, lung, and brain cancers also suggests a strong enrichment of high-ploidy cells among metastatic lesions as compared with the primary tumor (75, 76). As solid tumors progress, resources in the tumor become scarcer than the resources available in normal surrounding tissues (77). It is possible that these resource-poor environments push high-ploidy cells to leave the primary tumor into circulation and to thrive at locations with abundant access to nutrients (9).

Together, these findings suggest that resource availability in the TME manifests as differences in fitness between high- and low-ploidy cells, which go beyond replication and include invasion and persistence. Along with RNA and protein synthesis, DNA synthesis accounts for 50% of cellular ATP consumption in euploid cells (78, 79), and aneuploidy results in a net surplus of molecules from all three categories. Moreover, recent work suggests that the immune system may also play a role in conditioning the evolution of aneuploid cells (80, 81), further supporting the idea that aneuploidy is not always a favorable trait in tumor evolution. For all these reasons, the concepts highlighted in this perspective only begin to describe an emerging understanding of how tissue-specific environments condition ploidy. A significant amount of work is needed to understand the broad range of metabolic factors involved in coping with genomic instability and how tissue-specific resource availability compares to the demand of cancer cells of variable ploidies and their phenotypes, if demands are not met.

When put into context, a large body of data supports the idea that resource-poor tissue environments can seclude high-ploidy clones or prevent their evolution, and consequently can impede a tumor from coping with therapy-induced increases in SCNA rate. If this hypothesis is correct, the energetic costs of high ploidy can inform the minimum cytotoxic therapy dose needed to promote expansion of high-ploidy clones that a resource-poor environment ultimately cannot afford.

Authors' Disclosures

No disclosures were reported.

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